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RESEARCH ARTICLE

DDI-KGAT: A Graph Attention Network on Biomedical Knowledge Graph for the Prediction of Drug-Drug Interactions

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ABSTRACT Effective drug combination prediction is crucial for the success of drug discovery, but it is a challenging task due to drug-drug interactions and potential adverse drug reactions. In this work, a novel technique to DDI prediction using knowledge graph-based approach called KGAT is proposed, which utilizes attention mechanisms with graph convolution layers to capture important features and correlations between drugs and other entities such as targets and genes. Our model employs attention mechanisms to prioritize significant interactions and aggregates information through sum, mean, and max operations to enhance prediction accuracy. This allows KGAT to effectively mine high-order structures and semantic relationships within the knowledge graph. We evaluate our model on the KEGG dataset and compare its performance with existing state-of-the-art methods. The results show that KGAT outperforms these methods. Additionally, our approach has several advantages, including simplicity, interpretability, and low-dimensional complexity, making it a promising tool for accelerating drug discovery and development. By identifying novel drug combinations with improved efficacy and safety profiles, our approach has the potential to improve patient outcomes and support safer drug development. Our study highlights the potential of attention mechanisms in knowledge graph-based drug combination prediction, and we believe that KGAT can serve as a valuable framework for future research in this field.

INDEX TERMS Artificial intelligence, attention mechanisms, drug-drug interactions, graph neural networks, knowledge graph.

I. INTRODUCTION

When multiple drugs are taken together, they may interact and cause a change in the effectiveness of one or both drugs. This interaction is known as drug-drug interaction (DDI). DDIs can lead to adverse drug reactions (ADRs) [1], reduced drug efficacy, and even life-threatening conditions. ADRs are a significant concern in the field of healthcare, as they can lead to unexpected pharmacological effects and may even result in death. According to recent studies [2], the overall incidence of serious ADRs in hospitalized patients may be much higher than expected, with a fatality rate of 0.32% [3], making ADRs the fourth leading cause of death.

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The risk of DDIs increases with the number of drugs a person takes. The U.S. Centers for Disease Control and Prevention reports that over 10% of individuals consume five or more medications simultaneously, and the situation worsens as about 20% of older adults consume a minimum of ten medications [4]. This significantly elevates the possibility of experiencing ADRs. Therefore, effective identification of potential DDIs is critical in minimizing unexpected ADRs and maximizing treatment synergy to alleviate the impact of unexpected pharmacological effects [5].

Current DDI prediction techniques mainly integrate multiple databases to extract drug features, such as similarity features [6], adverse or side effects [7] and multi-task learning [8]. They make the assumption that drugs with similar representations will be having similar DDIs. In recent years,

machine learning (ML) techniques have become a promising approach to predict DDIs from various sources, such as electronic health records, scientific literature, and drug databases. However, most existing computational models using AI protocols often focus on combining multiple data sources and integrating prevalent embedding techniques [9], while paying less attention to potential correlations between drugs and other entities such as targets and genes. Although some recent studies have utilized knowledge graphs (KGs) for DDI prediction, these methods only learn node hidden embedding directly, limiting the ability to obtain comprehensive neighborhood details for every entity in the KG.

Therefore, this research aims to address these research gaps by leveraging comprehensive neighborhood details for every entity in KG and employing the knowledge graph attention network (KGAT), a technique that has not been previously used for DDI prediction despite its popularity in various applications. The objectives of this research are to demonstrate the effectiveness of KGs and graph neural networks (GNNs) in solving complex prediction tasks, to provide a simple and innovative approach that can potentially increase the prediction accuracy of DDIs with low-dimensional complexity, and to improve the accuracy and reliability of DDI prediction, potentially leading to better patient outcomes and safer drug development.

To achieve these objectives, the paper proposes a novel DDI-KGAT model that utilizes both attention mechanisms and sum aggregation layers to capture important features and enhance the performance of DDI prediction. The proposed model is assessed on Kyoto Encyclopedia of Genes and Genomes (KEGG) dataset, and the results are compared with state-of-the-art methods. By demonstrating its effectiveness, this research contributes to the field of DDI prediction by providing a simple and innovative streamlined solution. Additionally, the research highlights the importance of leveraging comprehensive neighborhood details for every entity in the KG and employing KGAT for DDI prediction.

Our research demonstrates that the integration of attention mechanisms and detailed entity relationships enhances prediction accuracy, establishing a significant advancement in the field. By accurately predicting potential DDIs, the DDI-KGAT model contributes to minimizing adverse drug reactions (ADRs), which are a major cause of morbidity and mortality, resulting in approximately 7,000 deaths annually [10]. While we do not directly measure reductions in morbidity and mortality, our model's improved DDI predictions hold clinical relevance, as demonstrated by a case study on Metformin and Dapagliflozin. This supports our techniques potential to enhance patient safety and inform safer drug development practices, ultimately contributing to better healthcare outcomes.

II. RELEVANT WORK

DDIs can have serious consequences for patients, including reduced therapeutic efficacy and increased risk of ADRs. Therefore, predicting potential DDIs is crucial for ensuring

patient safety. Various approaches have shown promise in predicting DDIs by leveraging various data sources. This review encompasses nine distinct techniques employed in DDI prediction, alongside a generalized KGAT approach applied to different domains as it has not been implemented for DDI prediction.

In the Similarity-Based category, Imming et al. [11] present an overview emphasizing the importance of understanding drug-target interactions. Zhang et al. [12] discuss the perspective of the FDA on predicting DDIs early in drug development. Gottlieb et al. [13] introduce the computational framework INDI for predicting drug interactions based on drug similarity measures and side effects. Vilar et al. [14] present a scalable methodology based on molecular structural similarity, offering a practical approach for large-scale analysis. Ferdousi et al. [15] introduced a computational method for predicting DDIs based on the functional similarity of drugs, utilizing key biological elements. Limitations of this category include challenges in capturing intricate relationships and dependencies between drugs and DDIs, as well as potential oversights in accounting for context-specific factors influencing Drug Interactions (DIs).

Classification-Based approaches, exemplified by Li et al. [16] conduct large-scale exploration using ML algorithms to predict DDIs based on chemical and biological properties. Shi et al. [17] propose LCM-DS, a novel approach using Dempster-Shafer theory of evidence for DDI prediction. Kastrin et al. [18] develop a statistical learning-based approach utilizing topological and semantic similarity features. Zhao et al. [19] introduces a novel RF network model for drug side effect prediction, leveraging the Mashup network embedding method to integrate heterogeneous drug and side effect networks. These approaches face challenges associated with the quality and quantity of available data, necessitating a substantial and diverse training set of known DDIs to achieve high accuracy. Moreover, there is a risk of overfitting to training data, potentially leading to poor generalization to new data.

Literature-Based techniques, as demonstrated by Tari et al. [20], use text-mining and reasoning to discover DDIs by analyzing drug metabolism properties. Tatonetti et al. [21] employ a data-driven approach using electronic health records to predict drug effects and interactions. Kolchinsky et al. [22] evaluate the performance of linear classifiers on articles containing pharmacokinetic evidence of DDIs. Challenges inherent in these techniques include reliance on the availability and accuracy of literature sources, which may be incomplete or biased. Additionally, there is the potential to overlook crucial information not explicitly mentioned in the literature, contributing to diminished accuracy owing to noise and variability in language use.

Using Link Prediction/Random Walk methodologies Cami et al. [23] propose a pharmacointeraction network model integrating multiple data sources for predicting unknown DDIs. Perozzi et al. [24] introduce DeepWalk, an online learning approach treating random walks as

sentences for learning node embeddings. Sridhar et al. [25] employ a probabilistic approach for predicting DDIs by modeling the collective similarity between drugs. Yan et al. [26] introduces IDNDDI, a novel computational method based on integrated drug feature similarity and node-based drug network diffusion. Challenges associated with these methodologies include limitations linked to the quality and completeness of network data. Moreover, there may be scalability issues when dealing with large networks, and the methods might fall short in capturing intricate interactions and dependencies between drugs and DDIs.

Under Neural Network approaches, Tang et al. [27] introduce LINE, a scalable approach for network embedding that preserves the first and second-order proximity of nodes. Wang et al. [28] present Structural Deep Network Embedding (SDNE), a deep autoencoder preserving both first and second-order proximity. Mrabah et al. [29] propose Variational Graph Autoencoder (VGAE), a model for node clustering that avoids feature twisting in the latent space. Limitations in these approaches include the need for a substantial amount of data for training. Furthermore, they are computationally intensive, demanding significant computing power and time for training. Additionally, they can pose difficulties in interpretation, making it challenging to understand how the model generates predictions.

For Ensemble-Based strategies, Zhang et al. [30] propose an ensemble-based approach for DDIs integrating various types of data and machine learning algorithms. Deepika and Geetha [31] present a meta-learning framework for DDI prediction using representation learning, involving popular algorithms like deep belief networks and autoencoders. The intricacy lies in the integration of multiple models, potentially necessitating substantial computational resources. Furthermore, there exists a vulnerability to overfitting if individual models within the ensemble exhibit high correlation.

Matrix Factorization techniques, illustrated by Belkin and Niyogi [32] introduce Laplacian eigenmaps for dimensionality reduction and data representation. Vilar et al. [33] propose a method for detecting potential DDIs based on interaction profile fingerprints. Cao et al. [34] present Grarep, a matrix factorization method leveraging global structural information. Shi et al. [35] introduce the Triple Matrix Factorization-based Unified Framework (TMFUF) innovatively predicts both conventional and comprehensive DDIs. These techniques face limitations in capturing intricate interactions and dependencies between drugs and DDIs, particularly struggling in performing well with sparse datasets. Furthermore, their effectiveness is contingent on the quality and completeness of the input data.

In Graph Embedding methods Ryu et al. [36] propose a deep learning approach for predicting drug-drug interactions using chemical and genomic data. Ma et al. [37] introduce an attentive multi-view graph autoencoder model for predicting DDIs. Liu et al. [38] propose a multi-modal deep autoencoder method for embedding structural networks to predict DDIs.

Feng et al. [39] introduces MFFGNN, a novel multi-type feature fusion model that effectively incorporating topological information from molecular graphs and interaction details between drugs. Wu et al. [40] provide survey of GNNs in data mining and machine learning, introducing a new taxonomy to categorize state-of-the-art GNNs. These methods encounter difficulties tied to the quality and completeness of the input data and graph structure, impeding their performance in large and complex networks. Additionally, they exhibit challenges in achieving optimal results in such intricate network settings.

Within KG DDI studies, Celebi et al. [41] evaluate different KG embedding approaches for DDI prediction using linked open data. Karim et al. [9] propose a DDI prediction method based on KG embeddings and a convolutional-LSTM network. Yu et al. [42] introduce SumGNN, a method for multi-typed DDI prediction via efficient KG summarization. Ren et al. [43] present BioDKG-DDI, a method for predicting DDIs based on drug KG fusing biochemical information. Extending this theme, Asada et al. [44] propose integrating heterogeneous knowledge graphs to enhance DDI extraction from literature, significantly improving performance on the DDIExtraction-2013 task. Introducing a novel approach, Deep Drug [45] presents a general graph-based deep learning framework for predicting DDIs and DTIs, demonstrating superior performance across various experiments and offering promising insights into the underlying biochemical relations. Further enriching this discussion, Luo et al. [46] systematically review DDI prediction methods that apply deep learning and knowledge graphs, highlighting the evolving use of these technologies in detecting potential DDIs more efficiently. These studies are constrained by the quality and completeness of the KG and associated data sources, leading to potential limitations in their effectiveness. They also demand high-quality data and may encounter scalability issues, particularly when dealing with extensive datasets. Additionally, there is a likelihood that these methods might not effectively capture novel or emerging DIs that are not explicitly represented in the KG.

Enveloped in KGAT (general) techniques, Wang et al. [47] propose KGAT, a knowledge graph attention network, demonstrating its effectiveness on real-world recommendation datasets. Cui et al. [48] present DETERRENT, a knowledge-guided graph attention network for detecting healthcare misinformation. Li et al. [49] introduce message-aware GAT for large-scale multi-robot path planning. Wu et al. [50] present DisenKGAT, a KG embedding approach with a disentangled graph attention network. Lan et al. [51] propose KGANCDA, a method for predicting circRNA-disease associations. Zhang et al. [52] develop a GAT with dynamic representation of relations for KG completion, demonstrating its effectiveness on benchmark datasets. KGAT methodologies may share limitations with KG approaches due to the introduction of additional attention mechanisms.

III. DEVELOPMENT OF KNOWLEDGE GRAPH-BASED MODEL FOR DDI PREDICTION

A. DATA COLLECTION AND PREPROCESSING

In the initial phase of data collection and preprocessing for DDI prediction, following steps are to be undertaken:

1) DDI EXTRACTION

Data is obtained from the public drug database (KEGG) to facilitate DDI extraction. The process involved parsing DDI information from an XML file and assembling an edge list of drug identifier combinations. This method ensures a comprehensive representation of DIs, which is crucial for accurate prediction and analysis.

2) KG CONSTRUCTION

The KG construction process comprised raw data retrieval from the KEGG dataset, conversion of data into an RDF graph using the Bio2RDF tool, and uploading the RDF graph to an RDF triplestore. It also involved the execution of federated SPARQL queries based on the billion triples benchmark to extract selected triples and the construction of the KG from the extracted triples, involving entities, relations, and entities in a triple structure.

Following the data collection and preprocessing steps, two primary data sources are to be created:

- A parsed DDI matrix containing drug-drug pairs.
- The constructed KG.

Due to the sensitive nature of drug data and the complexity of preprocessing steps involving proprietary tools and languages, we utilized a preprocessed KG by Lin et al. [53] to ensure reproducibility and facilitate accessibility for other researchers.

3) FEATURE EXTRACTION

In this step, we converted categorical data, such as drugs and entities, into numerical formats using embedding techniques. These embeddings, optimized during training, capture semantic relationships between drugs and their associated entities. Additionally, we applied graph-based methods to represent structural relationships within the DDI knowledge graph, employing neighborhood aggregation mechanisms inspired by GraphSAGE [54]. This approach enriched feature representations, enabling more accurate prediction of drug combinations.

4) MODEL DEVELOPMENT

We developed a KG based approach called DDI-KGAT for drug combination prediction using attention mechanisms. DDI-KGAT uses GNN with attention mechanisms to learn embeddings for the drugs and their interactions in the KG. The incorporation of attention mechanisms allow DDI-KGAT to capture important features and enhance the performance of drug combination prediction.

5) MODEL TRAINING AND EVALUATION

We trained DDI-KGAT model on a large dataset of drug combinations and evaluated its performance on the KEGG dataset. To ensure its robustness and prevent overfitting, we performed cross-validation. Next, we prepared the data and extracted features, splitting the dataset into training and testing sets. We defined the hyper-parameters, including embedding dimension, neighborhood sample size, l2_weight, learning rate, optimizer type, batch size, aggregator type, regularization strength, and number of epochs, before training the model using backpropagation and stochastic gradient descent (SGD) to minimize the loss function. We evaluated the performance of model on the testing set, using commonly used evaluation metrics such as accuracy (ACC) and F1 score, as well as Area under the Receiver Operating Characteristic Curve (AUC-ROC) and Area under the Precision-Recall Curve (AUPR). Furthermore, we visualized the predictions and the relationships of model between the drugs in the KG and compared its performance with other cutting-edge techniques for DDI prediction.

B. METHODOLOGY

Complete architecture of KGAT for predicting DDIs is as in Fig. 1, and is outlined below:

1) INPUT LAYER

The input layer consists of the KG containing entities such as drugs, targets, and genes, along with the relationships between them. It also includes DDI labels for binary classification, which are essential for training and evaluating the model's performance in predicting DDIs.

2) GRAPH EMBEDDING LAYER

In the DDI prediction task, we employ an Embedding Layer to map drugs and features in the KG to a low-dimensional vector space using embedding matrices. These matrices are trained during the training phase of model via backpropagation.

To encode drug nodes and their features, we adopt an entity embedding approach. Firstly, we create a dictionary encompassing all drugs and features present in the KG, assigning a unique index to each of them. Subsequently, we utilize these indices to construct a dense matrix representation of the KG.

To obtain entity embeddings, we utilize an embedding matrix that maps the indices to low-dimensional vectors. Each row of this matrix corresponds to a distinct entity in the KG. Throughout training, the embedding matrix is refined through backpropagation to minimize the loss function.

The Embedding Layer assumes a vital role in DDI-KGAT as it facilitates the representation of every KG entity as a compact vector. The vector representation captures the characteristics of the entity and its interconnectedness with other entities in the KG. These embeddings are then fed into the Attention Layer for further processing.

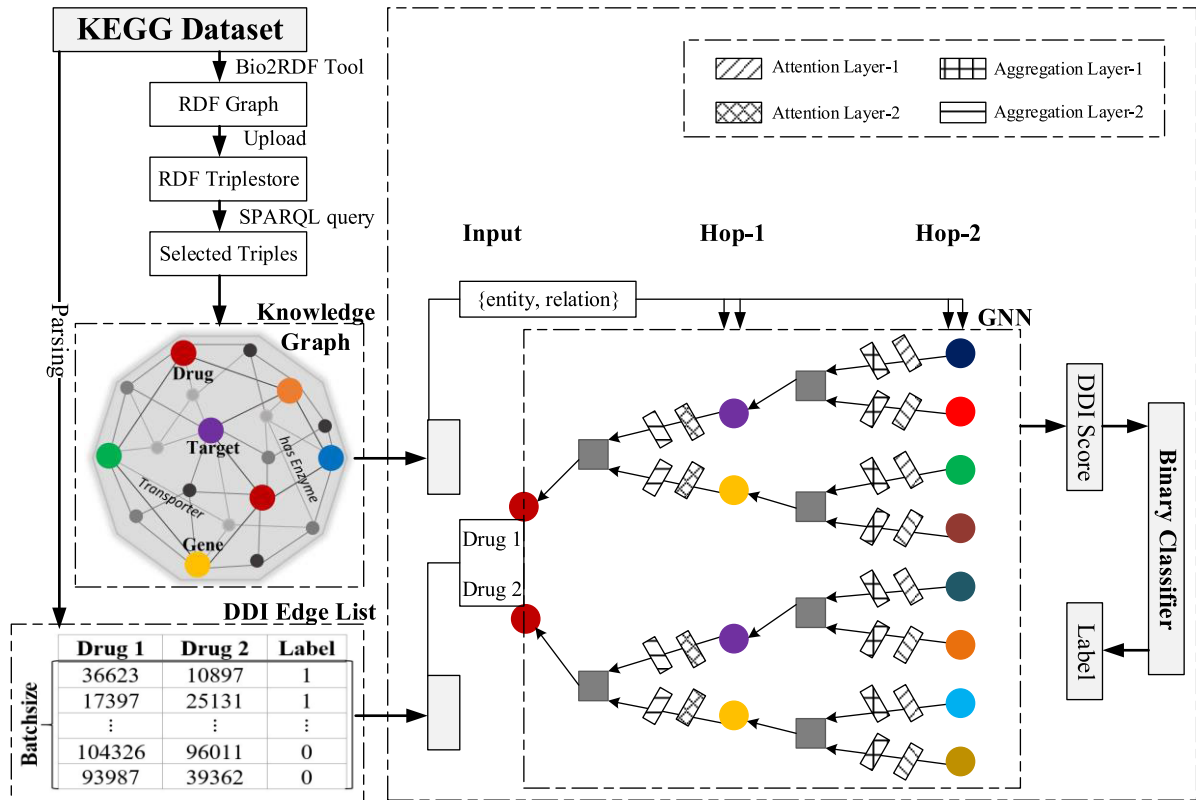


FIGURE 1. Overview of DDI-KGAT technique.

3) GRAPH ATTENTION LAYER

The Attention Layer in our project is a crucial component in capturing the relationships between entities in the KG and incorporating this information into the drug representation. The layer operates by computing attention weights for every neighbor of a drug node in the KG, which are a measure of the importance of every neighbor in the context of the drug node.

Leveraging a multi-head attention mechanism, our model adeptly prioritizes various subsets of neighbors, including drugs, diseases, genes, proteins, pathways, and interactions. By doing so, it identifies and emphasizes neighbors that hold substantial potential for interaction with the main drug, thus hinting at possible interactions. Furthermore, it keenly focuses on entities closely linked to the physiological effects of the main drug, such as genes or diseases it targets. This comprehensive approach not only enhances our understanding of drug interactions but also strengthens the fidelity of our drug representations within the KG. The attention weights (AW) are determined through a dot product between the drug node embedding and the embedding of each neighbor (ne), followed by the addition of the attention bias term (ab) and an attention weight tensor (aw). The resulting values undergo a softmax operation (σ) along the specified axis to generate a probability distribution over the neighbors. The mathematical representation is expressed as (1):

$$AW = \sigma((ne \cdot aw) + ab)_{axis=1}. \quad (1)$$

The attention weights are then used to obtain a context vector for the drug node by summing the weighted neighbor embeddings. The context vector captures the most important information from the neighbors in the KG and is used as a supplement to the original drug representation obtained from the Embedding Layer.

Overall, the Attention Layer is an essential component of current work as it allows the model to effectively capture the intricate connections among entities in the KG, which is crucial for accurately predicting DDIs.

4) GRAPH CONVOLUTION LAYER

The graph convolution layer is a key component in GNNs, including graph attention networks (GATs), for learning node representations in a graph. The goal of this layer is to aggregate information from a local neighborhood of a node and produce a new representation for the node that captures both the features of the node itself and the features of its neighbors.

There are different approaches to designing graph convolution layers, but a common one is to use a variant of the message passing scheme. The message passing scheme involves passing messages between nodes in a graph iteratively, with every message being a function of the features of the node itself and those of its neighbors. The messages are then aggregated across neighbors using a mean or summing operation, and the resulting aggregate message is used to update the representation of the node, see Fig. 2.

Our study explores the effectiveness of three distinct types of aggregators, including the sum, mean, and max aggregators, each influencing the aggregation process differently. In Fig. 2 and in the subsequent formulas (2), (3), and (4) we utilize the following terms summarized in Table. 1.

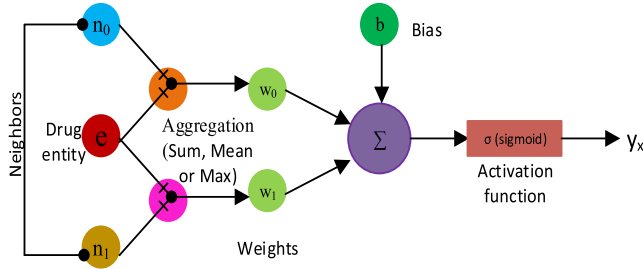


FIGURE 2. Structure of graph convolution layer.

TABLE 1. Notation definitions and descriptions.

Notation	Description
e	Drug Entity: Represents the intrinsic properties of the drug node in the KG.
n	Neighbor: Signifies the features of neighboring nodes that interact with the drug node.
y	Output: Denotes the resultant representation of the drug node after aggregation.
σ	Activation Function (Sigmoid): Introduces non-linearity to the output to help the model learn complex patterns.
$\cdot$	Dot Product: A mathematical operation used in the aggregation process to determine feature significance.
w	Weight Tensor: Responsible for weighting the importance of different features during aggregation.
b	Bias Term: Introduced to the weighted sum during aggregation to adjust the output.

a: Sum aggregation layer

In the Sum Aggregation Layer, the drug features and the context vector obtained from the Attention Layer are combined ($e + n$) to create a more comprehensive representation of the drug node in the KG. The drug features represent the intrinsic properties of the drug, while the context vector captures the influence of the neighbors of the drug in the KG. By weighting and combining these two sources of information, the model is able to capture both the intrinsic properties of the drug and its interactions with other entities in the KG, resulting in a more accurate prediction of DDIs.

$$y_{sum} = \sigma ((e + n) \cdot w + b). \quad (2)$$

b: Mean aggregation layer

In the Mean Aggregation Layer, the drug features and the context vector obtained from the Attention Layer are combined using the element-wise mean operation $((e + n)/2)$ to create a more balanced representation of the drug node in the KG.

This operation allows the model to capture both the individual properties of the drug and the overall characteristics of its neighbors in the KG, leading to a more comprehensive understanding of the drug and its potential interactions.

$$y_{mean} = \sigma (((e + n)/2) \cdot w + b). \quad (3)$$

It is essential to note that the element-wise operation is a common practice in graph convolution layers. This operation maintains simplicity and computational efficiency while still capturing essential information from both the drug node and its neighbors.

c: Max aggregation layer

In the Max Aggregation Layer, the drug features and the context vector obtained from the Attention Layer are combined using the element-wise maximum operation ($\max(e, n)$) to emphasize the most salient features of the drug node in the KG. This operation allows the model to focus on the most important properties of the drug and its neighbors, potentially leading to more accurate predictions of DDIs. However, the approach may also be more sensitive to outliers and noise in the data, and may require more careful tuning.

$$y_{max} = \sigma (\max (e, n) \cdot w + b). \quad (4)$$

5) OUTPUT LAYER

The FCNN is then used for binary classification, with the combined representation of the drug node as input. The FCNN is a type of neural network that consists of fully connected layers, which are connected in a sequence such that every neuron in a given layer is connected to every neuron in the subsequent layer. The output of the final layer of the FCNN is a single value that represents the predicted probability of a DDI between the two drugs. The binary cross-entropy loss function is used to train the model, which measures the difference between the predicted and actual labels for every training example and adjusts the weights of the model accordingly through gradient descent using the Adam optimizer.

6) LOSS FUNCTION

The loss function (L) is the binary cross entropy loss, which is a commonly used loss function for binary classification problems like DDI prediction. The binary cross entropy loss measures the difference between the predicted probability (p) of a DDI and the true label, which is either $y = 0$ (no DDI) or $y = 1$ (DDI present).

$$L = - [y * \log (p) + (1 - y) * \log (1 - p)]. \quad (5)$$

During training, model employs the Adam optimizer, a widely used algorithm that dynamically adjusts learning rates for each parameter based on gradients. The optimizer aims to find the best parameter configuration, minimizing the loss function and maximizing predictive performance on test data. By integrating a loss function and optimizer, the model

learns from errors, fine-tuning parameters for improved predictions. Evaluating the resulting predictions on a separate test set allows performance measurement and facilitates comparison with other models or baselines.

The DDI-KGAT technique consists of several graph convolutional layers followed by attention layers that learn the importance of different parts of the graph for the DDI prediction task. This allows the model to selectively focus on relevant parts of the graph while ignoring noise and irrelevant information. The final output layer combines the learned node embeddings with drug features to make a prediction about DDI.

Overall, the architecture of KGAT for DDI prediction involves multiple layers that are specifically designed to leverage the graph structure of the data and incorporate information from the graph in a principled way. The attention mechanism in the DDI-KGAT provides a flexible way to weight the contributions of different parts of the graph to the prediction task, making it a powerful approach for modeling complex relationships in the data.

IV. EXPERIMENTAL SETUP AND FINDINGS

A. EXPERIMENTAL SETUP

1) KEGG DATABASE

KEGG is a comprehensive database resource that integrates various biological information, such as genomic, chemical, and network information, to facilitate the analysis of complex biological systems. KEGG provides a vast collection of manually curated data on various biological pathways, diseases, and drugs, making it valuable for biomedical research. Dataset statistics are shown in Table 2. The KEGG dataset is widely used in the field of bioinformatics and has played a critical role in advancing our understanding of biological systems.

In this study, we utilized a combination of two data sources: a parsed DDI matrix and a constructed biomedical KG. The statistics of both data sources were analyzed to ensure data quality and accuracy.

TABLE 2. Basic statistics of the used kegg dataset.

Drugs	1925
DDIs	113965
Entities	129910
Relation	167
KG triplets	362870

2) HYPER-PARAMETER SETTINGS

After conducting a thorough case study and examining the effects of hyper parameters at various settings, we carefully tuned DDI-KGAT to the optimal configuration that yields best results for our task of DDI prediction.

We randomly divided all approved DDIs into positive samples and allocated them into training, validation, and testing sets using an 18/1/1 ratio. To ensure balance, we randomly sampled an equal number of negative samples from

the complement set of positive samples in all phases. We employed a random search and 8-fold cross-validation tests to optimize all trainable parameters using Adam. The training was set for 30 epochs with early stopping implemented, and we optimized other hyper-parameter settings for the validation set, which are provided in a Table. 3 for reference.

TABLE 3. Hyper-parameter settings.

Hyper-parameter	Value
Neighbor sampling size	16
Dimension of embedding	32
Depth of receptive field	2
Batch size	2048
L2 regularizer weight	2e-8
Learning rate	5e-3

3) BLOCK DIAGRAM OF EXPERIMENTAL WORK

The block diagram in Fig. 3 illustrates the main experimental workflow of this research, from data collection and DDI extraction through to model development, training, and evaluation. This structured approach ensures a thorough and systematic methodology for predicting DDIs.

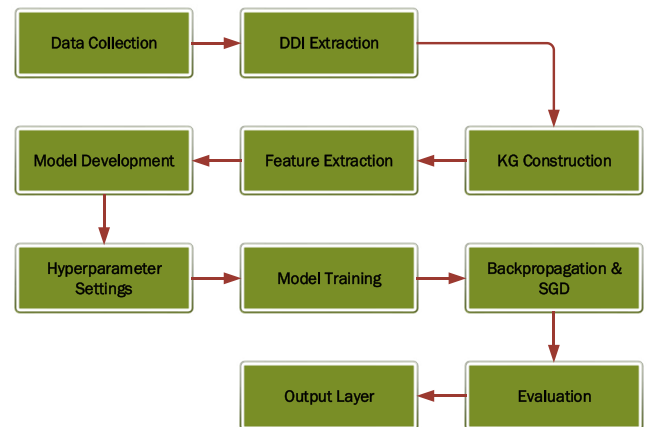


FIGURE 3. Block diagram of the experimental workflow.

B. RESULTS

In this section, we present the results of experiments evaluating the performance of the proposed DDI-KGAT model for predicting DDIs on the KEGG dataset. We categorize the results based on the performance of three different types of aggregators commonly used in GNNs: sum, mean, and max. Each aggregator is evaluated separately, and their respective results are compared. Additionally, we provide insights into the interplay between the drugs, targets, genes, and neighboring entities.

1) PERFORMANCE WITH SUM AGGREGATOR

We first examine the performance of our model using the sum aggregator. The results as in Fig. 4, show that the DDI-KGAT achieves the highest average AUC-ROC of 0.9447, ACC of

0.8901, F1 score of 0.8917 and AUPR of 0.9255 outperforming the state-of-the-art approaches.

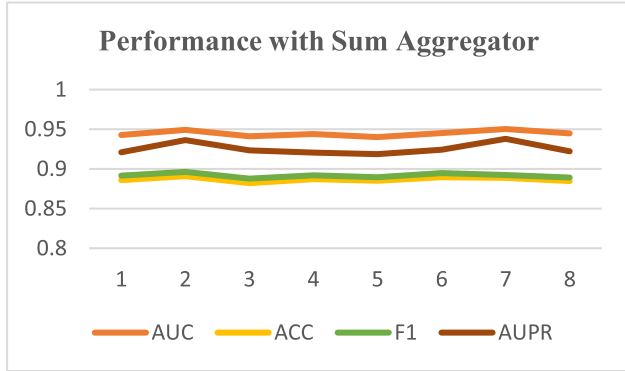


FIGURE 4. 8-fold results of DDI-KGAT with sum aggregator.

2) PERFORMANCE WITH MEAN AGGREGATOR

Next, in Fig. 5 we evaluate the performance of this strategy using the mean aggregator. The results demonstrate that the DDI-KGAT achieves the average AUC-ROC of 0.9419, ACC of 0.8853, F1-score of 0.8867 and AUPR of 0.9226, which is comparable to the performance achieved with the sum aggregator.

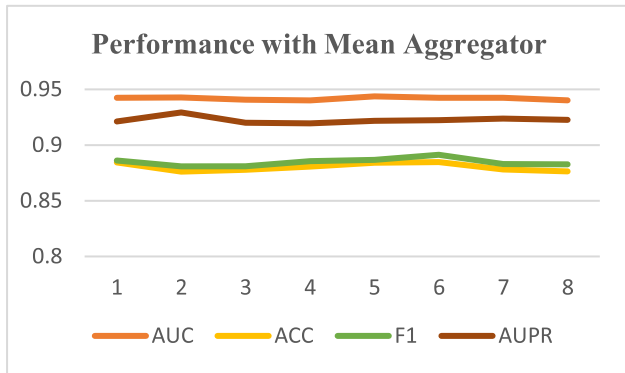


FIGURE 5. 8-fold results of DDI-KGAT with mean aggregator.

3) PERFORMANCE WITH MAX AGGREGATOR

We then examine the performance of our technique using the max aggregator in Fig. 6. The results indicate that the DDI-KGAT technique achieves the average AUC-ROC of 0.9397, ACC of 0.8799, F1-score of 0.8816 and AUPR of 0.9207, which is slightly lower than the performance achieved with the sum aggregator.

4) COMPARISON OF AGGREGATORS

Finally, we present a comparison of the performance of DDI-KGAT in Fig. 7, using the three different types of aggregators. The results show that the sum aggregator outperforms both the mean and max aggregators in terms of AUC-ROC, ACC, F1-score and AUPR. Findings suggest that the sum aggregator is the most effective aggregation function for the proposed DDI-KGAT model in the task of predicting DDIs.

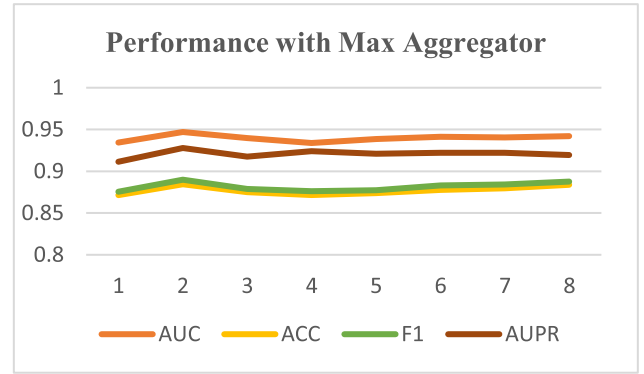


FIGURE 6. 8-fold results of DDI-KGAT with max aggregator.

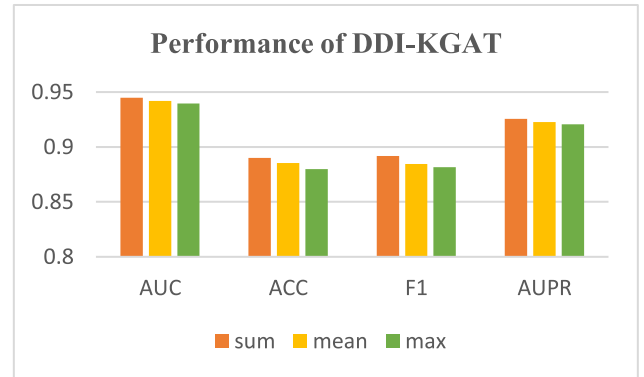


FIGURE 7. DDI-KGAT overall results.

By organizing the results into specific topics based on the aggregators, we provide a clearer understanding of the performance of the DDI-KGAT model and its interplay with different types of aggregators. These insights contribute to the understanding of the relationships between drugs, targets, genes, and neighboring entities, enhancing the significance of our findings in predicting DDIs.

V. DISCUSSION AND CONCLUSION

A. DISCUSSION

GNNs have emerged as powerful tools for DDI prediction using KGs. Aggregators play a crucial role in these models as they determine how neighboring nodes are combined to update the embedding of a central node. The choice of aggregator depends on the specific task and the characteristics of the graph being used. In GNNs, there are several common types of aggregators. We have used: sum, mean, and max aggregators. Each aggregator has its own characteristics in terms of computational efficiency and memory usage.

The sum aggregator is widely used in GNNs and calculates the sum of the embeddings of neighboring nodes. While it is computationally efficient, it requires more memory as all the outcomes are first saved and then summed before being passed to the next step. However, the sum aggregator may be sensitive to outliers and may not always capture the most relevant information from neighboring nodes. On the other hand, the mean aggregator computes the mean of the embeddings of neighboring nodes. It uses less memory as the mean is computed at each step without the need to save previous

values. This temporary usage of memory during the aggregation process contributes to its low memory requirements. However, the mean aggregator is not as computationally efficient as the sum aggregator. The max aggregator computes the maximum value among the embeddings of neighboring nodes. Similar to the sum aggregator, it requires more memory as all outcomes are saved and a comparison is drawn to obtain the maximum value. While the max aggregator has shown good results in many works, it may not perform well for DDI prediction, as our research has indicated. We found that the max aggregator does not yield satisfactory results in this context, delivering useful observations for researchers in the field.

Considering memory usage and computational power, the sum aggregator is more memory-efficient and requires fewer computational resources compared to the mean and max aggregators. It also outperformed the mean and max aggregators in predicting DDIs, as it effectively captures the most relevant information from neighboring nodes. Moreover, the sum aggregator is easy to implement and interpret, making it a popular choice for various applications. In summary, the sum aggregator in GNNs is a reliable and efficient function that can effectively update the embeddings of central nodes, providing valuable insights for predicting DDIs and other applications.

1) COMPARISON WITH SIMILAR WORK ON PREDICTING DDIs USING KNOWLEDGE GRAPH NEURAL NETWORK (KGNN) ON KEGG DATASET

This research study has shown that incorporating an attention mechanism with a simpler aggregator has yielded even better results than the concatenate aggregator, which was claimed to be the best so far (in predicting DDIs using KGNN on KEGG dataset) by Lin et al. [53].

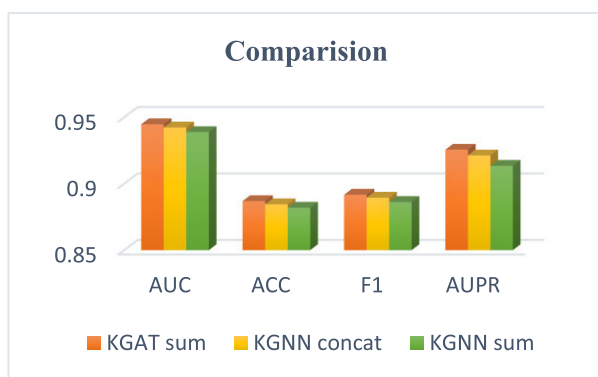


FIGURE 8. DDI-KGAT (sum) vs KGNN (concat, sum).

The finding in Fig. 8, suggests that DDI-KGAT has surpassed this state-of-the-art approach and Table. 4 shows the achieved highest ACC, F1-score, AUC-ROC, and AUPR values for the DDI prediction task compared to KGNN technique.

The results of DDI-KGAT as in Fig. 9 have significant implications for the field of KG embedding, especially in domains where high accuracy and low-dimensional

TABLE 4. Comparison of DDI-KGATx vs KGNNx technique.

Methods Metrics	KGNNx			DDI-KGATx		
	neighbor	sum	concat	max	mean	sum
AUC-ROC	0.9295	0.9387	0.9422	0.9397	0.9419	0.9447
ACC	0.8681	0.8818	0.8844	0.8799	0.8853	0.8901
F1 score	0.8726	0.8862	0.8894	0.8816	0.8867	0.8917
AUPR	0.9052	0.9134	0.9212	0.9207	0.9226	0.9255

complexity are essential, such as drug discovery. By combining the attention mechanism with a simpler aggregator, our technique has not only improved the performance of the KGNN model [55] but has also reduced its complexity, making it more computationally efficient.

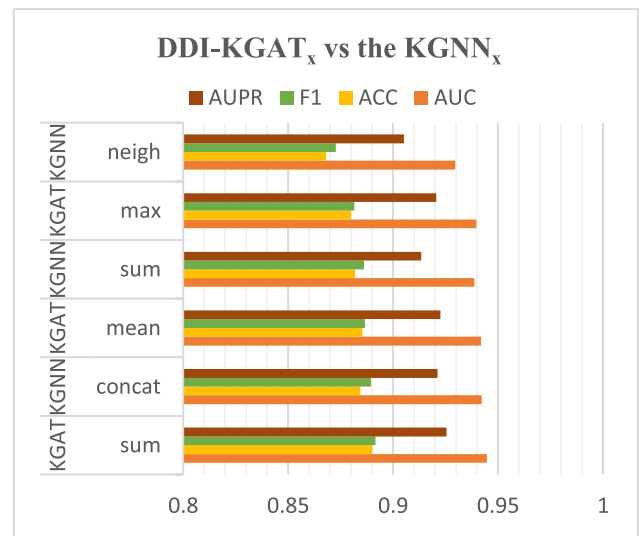


FIGURE 9. Comparison of DDI-KGATx vs KGNNx technique.

2) COMPARISON WITH STATE-OF-THE-ART TECHNIQUES USING KNOWLEDGE GRAPH FOR PREDICTING DDIs

In the study by Feng et al. [39] on DPDDI, deep learning frameworks are employed; however, a limitation arises in terms of interpretability, making it challenging to comprehend the underlying factors influencing predictions. Yu et al. [42] propose SumGNN, which focuses on efficient KG summarization for multi-typed drug interaction prediction. Nonetheless, the method encounter difficulties in capturing intricate relationships within the KG, potentially impacting prediction accuracy. Ren et al. [43] introduce BioDKG-DDI, which combines biochemical information with a drug KG. Nevertheless, the integration of multiple data sources presents complexities and potential challenges in handling heterogeneous data. SA-DDI technique by Yang et al. [56] include challenges in interpreting the identified substructures, requiring further investigation for meaningful interpretation. Lastly, Hao et al. [57] utilize a three-way decision and stack multiple complex techniques, which can lead to increased computational complexity and

hinder scalability. Although their performance is comparable to DDI-KGAT as in Fig. 10, but it is not offering the same level of simplicity and computational efficiency.

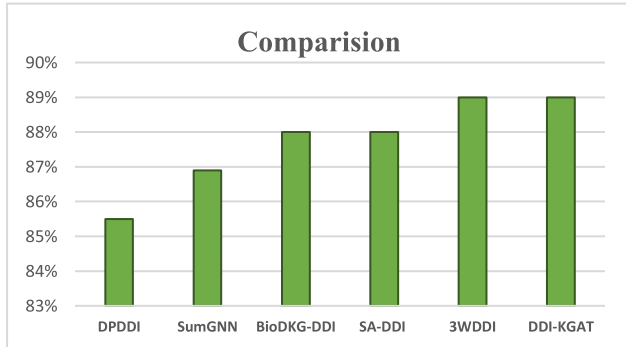


FIGURE 10. Comparison of DDI-KGAT vs State-of-the-art techniques.

The effectiveness of present strategies is validated by the outstanding F1 score, accurately capturing true positives and reducing false positives in DDI prediction. Prioritizing this metric allows for meaningful comparisons with existing literature, offering a comprehensive assessment of precision and recall. While AUPR, ACC, and AUC provide insights into classifier performance, the emphasis on the F1 score highlights the ability of the current technique to strike a balance between precision and recall. This focus underscores our commitment to addressing DDI prediction challenges and establishes a new benchmark in the field. In conclusion, the F1 score serves as a reliable metric, demonstrating the effectiveness of DDI-KGAT in predicting DDIs.

TABLE 5. Comparison of DDI-KGAT vs state-of-the-art techniques.

Methods	F1-score (%)
DPDDI (Feng et al., 2020) [39]	85.5
SumGNN (Yu et al., 2021) [42]	86.9
BioDKG-DDI (Ren et al., 2022) [43]	88
SA-DDI (Yang et al., 2022) [56]	88
3WDDI (Hao et al., 2023) [57]	89
DDI-KGAT (Ours)	89

Our analysis given in Table. 5, reveals that DDI-KGAT stands out from the aforementioned publications due to its exceptional performance, high accuracy, low-dimensional complexity, and computational efficiency. The approach offers a simpler and more interpretable solution, while still achieving comparable or even superior results, demonstrating its effectiveness and practicality in the context of DDI prediction. Additionally, DDI-KGAT demonstrates superior computational efficiency, scalability, and integration ease compared to existing methods. It requires fewer computational resources and offers faster inference times due to its optimized architecture, maintaining robust performance with increasing data volumes and knowledge graph complexity. Its interpretable predictions and user-friendly interface facilitate seamless integration into clinical workflows, making it the preferred choice for DDI prediction in real-world settings. This positions DDI-KGAT as a benchmark in the field, setting

new standards for efficiency and usability in DDI prediction. See the Table. 6 for a summary evaluation of ours compared to state-of-the-art techniques.

TABLE 6. Evaluation of DDI-KGAT vs state-of-the-art techniques.

Methods	Computational Efficiency	Scalability	Integration Ease
DPDDI	Moderate to High	Limited	Moderate
SumGNN	Moderate	Moderate	Limited
BioDKG-DDI	Moderate	Moderate	Limited
SA-DDI	Moderate to High	Limited	Moderate
3WDDI	Low to Moderate	Limited	Limited
DDI-KGAT (Ours)	High	High	Easy

B. CASE STUDY

1) EVALUATING DDI PREDICTIONS FOR METFORMIN AND DAPAGLIFLOZIN

To demonstrate the clinical relevance and interpretability of our DDI-KGAT model, we conducted a detailed case study focused on the predicted interaction between Metformin, a widely used antidiabetic drug, and Dapagliflozin, an SGLT2 inhibitor. Both drugs are commonly prescribed to manage type 2 diabetes but their interaction potential is not extensively documented in clinical practice.

Our DDI-KGAT model predicted a significant interaction between Metformin and Dapagliflozin. This interaction could potentially enhance the hypoglycemic effects when used concomitantly, thereby increasing the risk of lactic acidosis [58], a rare but serious complication that involves the enhanced glucose regulatory effect. The binary output of the model indicated a “positive” interaction, suggesting heightened attention when these drugs are prescribed together.

C. CONCLUSION

In summary, this study demonstrates the effectiveness of integrating KGs and GNNs for DDI prediction, with a particular focus on leveraging attention mechanisms and sum aggregation in the DDI-KGAT model. Our results showed a significant improvement in prediction accuracy, computational efficiency, and dimensionality reduction compared to traditional methods. By capturing intricate neighborhood details within the knowledge graph, the DDI-KGAT model effectively predicted potential DDIs, minimizing the risk of ADRs.

Furthermore, the applicability of the DDI-KGAT model extends to various drug types and populations, as its architecture is designed to generalize across different drug interactions, including unseen combinations. The model’s ability to adapt to diverse datasets suggests its scalability for use in larger healthcare systems, enabling more accurate and interpretable DDI predictions that could significantly improve patient outcomes and assist in the development of safer drugs. These findings set a strong foundation for future research in KG embedding, especially in high-stakes domains like

healthcare, where accuracy, efficiency, and scalability are critical.

D. LIMITATIONS AND FUTURE WORK

Efforts should focus on enhancing the accessibility and transparency of the preprocessing pipeline by developing user-friendly, open-source tools to promote broader adoption and reproducibility. Future research could refine the model architecture to better handle specific or complex drug interactions, such as those involving rare or less-studied drugs. Additionally, incorporating domain-specific knowledge or integrating additional data sources could address current limitations and advance the applicability of our findings.

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